Module 12 Lecture - Complex Experiments

Research Methods in Psychology

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1 Chapter Overview

1.1 Chapter Overview

- Up until now, we have only talked about experimental designs that deal with _____ manipulated/independent variable and one measured/dependent variable.

? Prior to experiments, we dealt with designs only focusing on measured variables, what were those called?

- (●) A) Correlational
- ( ) B) Observational
- ( ) C) Probabilistic
- ( ) D) Multivariate

Explanation:

- However, we have designs that can look at two (or more) IVs at once and see their individual and _____ impact on the DV!
- We refer to these as _____ designs, and they can be helpful in unpacking nuance in certain relationships.

2 Experiments with Two IVs

2.1 Overview

- We can add a second (and third) _____ variable if we are curious about more than one.
- In addition to the individual effects of both of the IVs, we also get an _____ effect that describes how they change each other's relationship with the outcome.

? We still have to be concerned about causality, because this is still an experiment - what validity is that most related to?

- (●) A) External
- ( ) B) Statistical
- ( ) C) Internal
- ( ) D) Construct

Explanation:

- Statistically, we might say this interaction is a "_____ in differences"
  - Practically, this means that the differences between our groups may be different based on some other trait.
  - More on this later

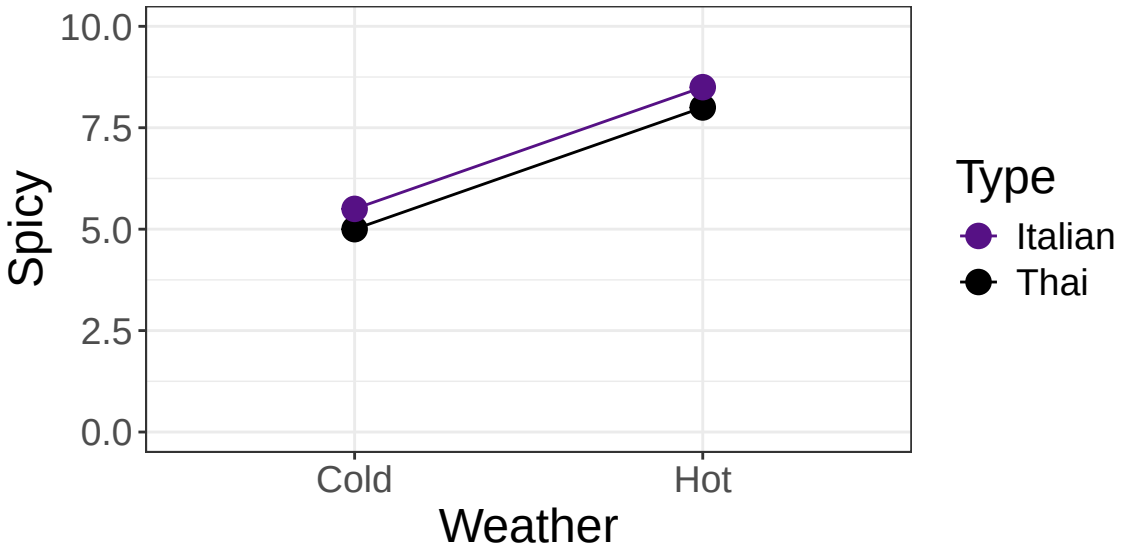
2.2 Intuitive Interactions

- When confronted with a causal relationship, sometimes we might say, "well it _____" - what it depends on is the second (or third) IV

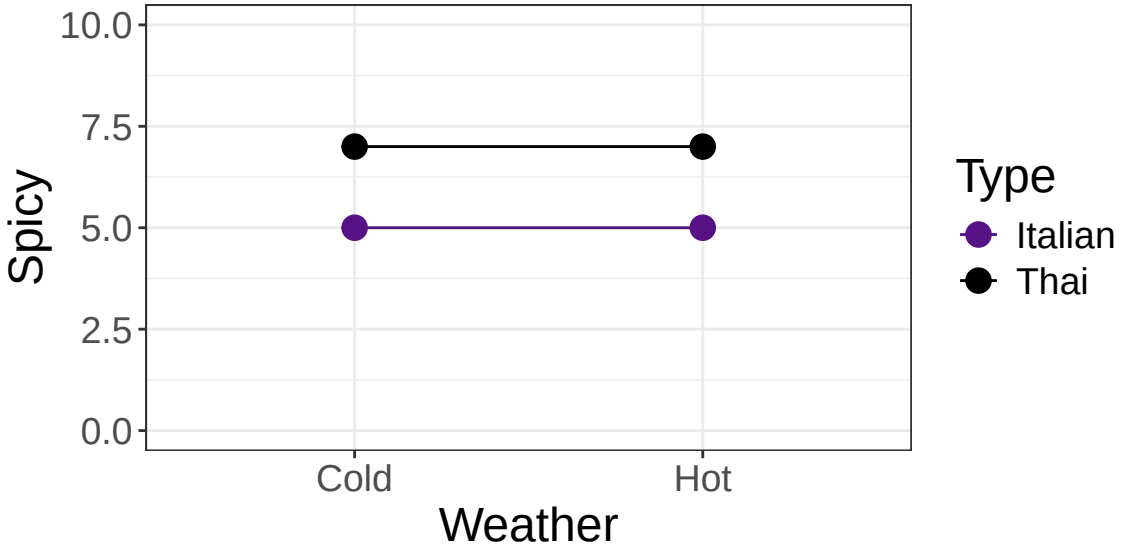
Discuss: Previously, we described two types of variables that play a role in the casual link between two others. What were they and describe each? (hint: they both start with the letter M)

- We can see this even in our personal experiences, and many relationships do depend on _____ factors
- *Example:* I am assessing how spicy I like my food (on a scale of 1 to 10; my outcome). First, is it cold or hot outside (IV 1)? Second, am I eating Thai or Italian (IV 2)? It is possible that my answer will be different based upon both of the IVs.
- **4 Possible Outcomes:**

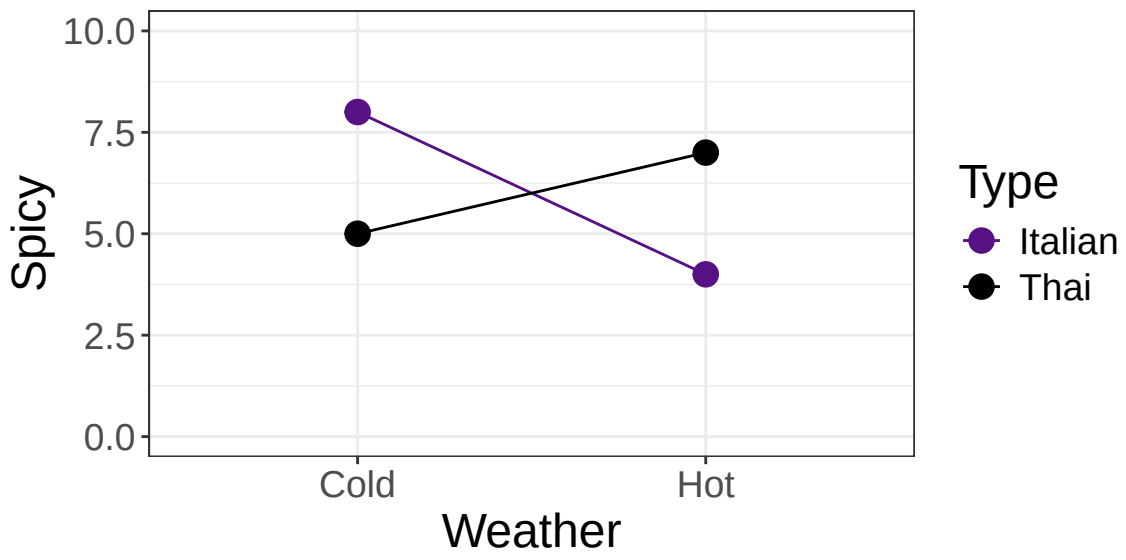
- I like all of my food spicier when it is hot - Weather effect, but *not* food



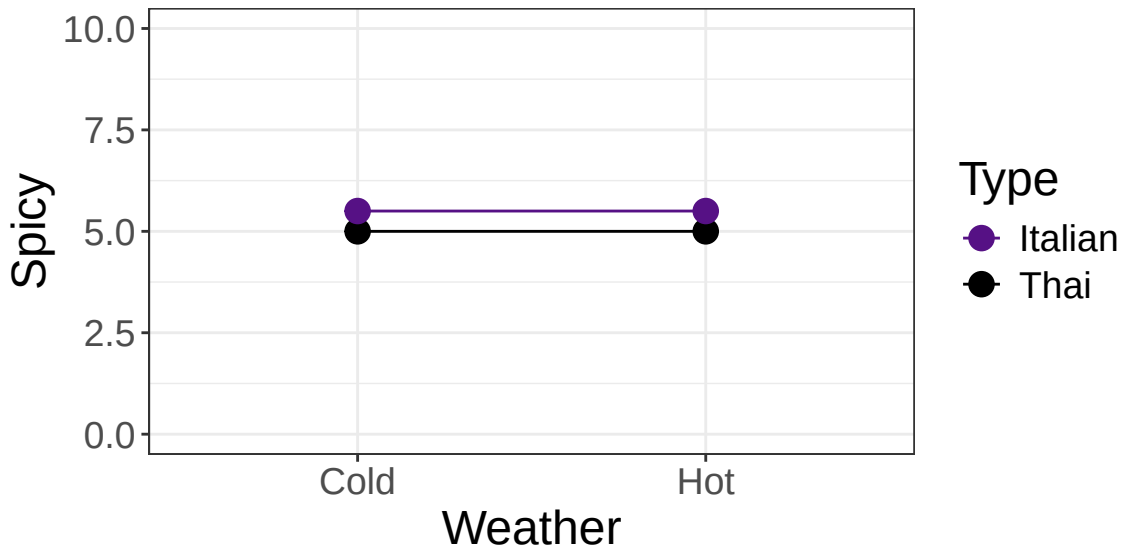
- I like Thai food spicier that Italian, regardless of weather - Food effect, but *not* temperature



- Whether I like by food spicy or not depends on both the weather, and type of food - interaction effect
- Specifically, we are looking to see whether we have a _____ interaction, like in the graph below:



- My preference for spice doesn't change, regardless of food type or weather - null findings



2.3 Study Two IVs

- When we work with _____ than one IV, we use a factorial design.
- This creates more _____ unique conditions = # of Conditions in IV 1 x # of Conditions in IV 2 = total number of conditions
- Both IVs do *not* have to be _____. Often, one will be some categorical, measured trait (e.g., gender, ethnicity, etc.)

  - Sometimes these get called "quasi-manipulated" because they are not actually manipulated but we _____ them as such.

- In addition to our statistics, we should show these differences in _____! Interaction effects become especially clear with visual evidence.

2.4 Limit Testing

- Factorial designs can help us find whether outcomes are different for different _____ of people, or maybe an intervention only works if another intervention is present
- A strong intervention may not be as effective in a _____ group of people.

? Review - When it comes to a 'strong' intervention what aspect of statistical validity will be mostly looking at?

- (●) A) Precision
- ( ) B) Significance
- ( ) C) Reproducability
- ( ) D) Effect Size

Explanation:

- This can be a _____ to our external validity, as we demonstrate findings in a more heterogeneous group.
- We also can establish whether one variable appears to _____ another on the relationship with the outcome variable.

2.5 Test Theories

- For some _____ reasons, we may have good reason to believe that an effect differs based on some demographic variable.
- *Example:* I have a new intervention meant to encourage flexibility in learning and taking in new content. However, I recognize that the neuroplasticity of

older adults is just lesser in general. Therefore, I believe my intervention will likely be more effective for younger adults, than it will for older adults.

- In essence, we may be able to add _____ and "it depends" to our hypotheses and investigate with factorial designs.

Discuss: Try coming up with another example of a situation where you might find an effect 'depends' on something else? (Maybe something in your own proposal)

2.6 Main Effects & Interactions

- **Main Effects** are those that come from each IV on the outcome.
  - The main effect is _____ as an average over the levels of the other IV. Similar to how we "control" for other variable in multiple regression.
  - You have 1 main effect for each IV
- **Marginal Means** are the _____ that we use to determine whether a main effect is present
  - We can test significance by taking the difference of the two marginal means, and calculating 95% CIs. If CIs _____ 0 $\rightarrow$ non-significant

DV: Reaction time (ms)		IV_1: Photo type		
		Alcohol	**Plant**	Main effect for IV_2: Word type
IV_2: Word type	**Aggressive**	**551**	**559**	**555 (average of 551 and 559)**
	Neutral	**562**	**552**	**557 (average of 562 and 552)**
Main effect for IV_1: Photo type		**556.5 (average of 551 and 562)**	**555.5 (average of 559 and 552)**	

- An **interaction effect** can be detected by looking at the differences of the main effect differences. If they are _____ different from one another, then we would say that there is an interaction effect
  - Interactions are often treated as _____ important, theoretically, that main effects - when they are significant.
- Conventional wisdom: If interaction is _____, focus on that mostly. If interaction is non-significant, focus on main effects of IVs.
  - Interpreting the main effects with a significant interaction can be leaving out _____ information!

Discuss: Try explaining, in your own words, why you suspect only looking at the main effects is a bad idea when you have a significant interaction?

- *Stats sidebar:* This type of analysis is usually done via Two-way ANOVA, which does all the work of _____ significance of interactions, and main effects for us.

3 Factorial Variations

3.1 Overview

- Just like with other experiments, we can lay out a factorial design as being between-groups or b58fc729-690b-4000-b19f-365a4093b2ff-7B7B3C20626C616E6B2077697468696E203E7
- But, we can designate _____ variable as between or within, leading to a total of 3 possible designs:
  - Independent-Groups Factorial
  - Within-Groups Factorial
  - _____ Factorial

3.2 Independent-Groups Design

- This is when _____ IVs are between-groups (i.e., participants are arranged into entirely separate groups)
- One nuance is that this will likely require the largest sample _____, as each group will have about 1/4th the total number of participants

Discuss: What validity does a smaller sample size primarily hurt. Explain why this happens

- *Example:* I measure whether honors or regular students have differential benefits when they are place in living-learning or regular dormitories

3.3 Within-Groups Design

- Much like with previous within-groups designs, this is when participants see _____ possible condition.
- One thing to watch out for is the need for _____ to prevent order effects

  - Think about how many _____ of condition orders you may need!

- *Example:* I am interested in seeing whether a certain note-taking strategy and a review strategy help performance on a test - so I have the same people counterbalances to different combinations of both conditions.

3.4 Mixed Factorial Design

- This is when one IV is b58fc729-690b-4000-b19f-365a4093b2ff-7B7B3C20626C616E6B206265747765656E2
and the other is within-group.
- This is fairly common if we have one _____ variable (between-groups) and one manipulated variable that both demographics are exposed to each level (within-groups).
- *Example:* I am determining whether first-generation college students or legacy students benefit more from a mentoring program, in terms of confidence. Both first-gen and legacy students have a period that they are exposed to the mentoring program and a time period without.

3.5 More Conditions

- Many _____ are going to naturally have more than one level
  - E.g., race, ethnicity, gender, etc.
- We can use these in factorial designs all the same - and we write it as: AxB Design.
  - Where A = Number of conditions in IV 1
  - Where B = Number of conditions in IV 2
- Statistics here get more _____ to interpret - but a good starting point is to try to use a line plot just like what we have done previously and see if lines cross or are parallel.

3.6 More IVs

- Prof. Paul Moes: "God himself cannot interpret a 4-way interaction - neither can you"
- We can do 3 IVs, but with each _____ variable the interpretation becomes exponentially more difficult and complicated.
  - One popular alternative is to do this as a multiple _____ model instead

  - *Stat sidebar:* ANOVA and linear regression are both types of the general _____ model, so, in a roundabout way, these are actually equivalent!

- Remember to think carefully about what sorts of conclusions you can draw with a design before you use it, and whether an alternative provides a more simpler conclusion.

? Review - we previously emphasized the importance of keeping conclusions as simple as possible, what vocab word captures that sentiment?

- (●) A) Pattern
- ( ) B) Purpose
- ( ) C) Parsimony
- ( ) D) Promise

Explanation:

4 Identify Factorial Designs

4.1 Reading Empirical Articles

- Look for words like …
  - "_____ ANOVA"
  - "Factorial"
  - "Interaction" or "Main Effects"
- You may also see phrasing like "2 x 2 design", referring to the two conditions of each IV.
- Sometimes a multiple _____ model and a factorial design can be described in somewhat similar conclusions - you might have to work through the results to figure out which one was used

4.2 In Popular Media

- Look for _____ like …
  - "_____"

  - “Only when”
- You may also look for demographic _____ …
  - “For males this was the results, but for females…”

The instructor-provided glossary may not include all terms worth memorizing, make sure you consider using the vocabulary list in you book and your own judgment to make sure you have all relevant terms